Light

GCSE Computer Science - Cyber Security Prevention (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the primary function of a firewall?

- ○ **A.** To monitor and control incoming and outgoing network traffic based on security rules
- ○ **B.** To scan files for viruses
- ○ **C.** To encrypt all data on the hard drive
- ○ **D.** To create backup copies of files

Q2 (1 mark)

What does anti-malware software do?

- ○ **A.** Blocks all internet connections
- ○ **B.** Detects, prevents, and removes malicious software from computer systems
- ○ **C.** Manages user passwords
- ○ **D.** Controls physical access to buildings

Q3 (1 mark)

What is penetration testing?

- ○ **A.** Testing how deep water can penetrate a computer case
- ○ **B.** Testing network cable lengths
- ○ **C.** Authorised simulated attacks on systems to identify security vulnerabilities before malicious attackers can exploit them
- ○ **D.** Testing how quickly users can type passwords

Q4 (1 mark)

What is encryption?

- ○ **A.** Converting data into a coded form that can only be read with the correct key
- ○ **B.** Deleting files permanently
- ○ **C.** Compressing files to save storage space

- ○ **D.** Creating multiple copies of data

Q5 (1 mark)

What are user access levels?

- ○ **A.** The physical height at which monitors are placed
- ○ **B.** The number of times a user can log in per day
- ○ **C.** The speed of the user's internet connection
- ○ **D.** Different permission settings that control what resources and actions each user can access

Q6 (1 mark)

Which of the following is an example of physical security?

- ○ **A.** Installing antivirus software
- ○ **B.** Using strong passwords
- ○ **C.** Locking server rooms and using security cameras
- ○ **D.** Enabling a firewall

Q7 (1 mark)

Which password characteristic most increases security?

- ○ **A.** Using your name so you remember it easily

- ○ **B.** Using the same password everywhere for convenience
- ○ **C.** Using a long combination of uppercase, lowercase, numbers, and symbols
- ○ **D.** Writing the password on a sticky note near the computer

Q8 (1 mark)

What does a firewall examine when deciding whether to allow network traffic?

- ○ **A.** The source address, destination address, and port numbers of data packets
- ○ **B.** The colour of the network cables
- ○ **C.** The brand of the computer sending data
- ○ **D.** The age of the files being transferred

Q9 (1 mark)

Why must anti-malware software be regularly updated?

- ○ **A.** To change the software's colour scheme
- ○ **B.** To recognise newly created malware through updated signature databases
- ○ **C.** To make the computer run faster
- ○ **D.** To increase the size of the hard drive

Q10 (1 mark)

What threat does encryption primarily protect against?

- ○ **A.** Unauthorised reading of intercepted data
- ○ **B.** Power failures
- ○ **C.** Computer overheating
- ○ **D.** Software crashes

Q11 (1 mark)

What is the 'principle of least privilege'?

- ○ **A.** Giving all users maximum access to simplify administration
- ○ **B.** Removing all privileges from all users
- ○ **C.** Only giving users the minimum access rights needed to perform their job
- ○ **D.** Privileged users should use the system least often

Q12 (1 mark)

What is biometric security?

- ○ **A.** Security based on the user's biography
- ○ **B.** Security using physical characteristics like fingerprints or facial recognition to verify identity

- ○ **C.** Security using only numeric passwords
- ○ **D.** Security that protects biological research

Q13 (2 marks)

A company hires security professionals to attempt to break into their network and report any weaknesses found.
What is the main benefit of this approach?

- ○ **A.** It identifies vulnerabilities that can be fixed before real attackers exploit them, and tests whether existing security measures are effective
- ○ **B.** It guarantees the network can never be hacked
- ○ **C.** It permanently blocks all future attacks
- ○ **D.** It removes the need for other security measures

Q14 (2 marks)

A school network administrator configures the firewall to block all incoming traffic on port 22 (SSH) from external networks.
What is the purpose of this configuration?

- ○ **A.** To speed up the school's internet connection
- ○ **B.** To allow more students to connect simultaneously
- ○ **C.** To improve the quality of video streaming
- ○ **D.** To prevent external users from remotely accessing school servers via SSH, reducing the attack surface

Q15 (2 marks)

A business requires employees to use a VPN (Virtual Private Network) when working remotely.
How does a VPN protect data?

- ○ **A.** It makes the internet connection faster
- ○ **B.** It creates an encrypted tunnel between the employee's device and the company network, protecting data from interception
- ○ **C.** It blocks all websites except work-related ones
- ○ **D.** It automatically backs up all files to the cloud

Q16 (2 marks)

A hospital implements different access levels: doctors can view and edit patient records, nurses can view records but not edit diagnoses, and receptionists can only view appointment schedules.
What security principle does this demonstrate?

- ○ **A.** Role-based access control - permissions are assigned based on job requirements, limiting each role to necessary access only
- ○ **B.** Maximum access - everyone should see everything for efficiency
- ○ **C.** No access - nobody should access patient data
- ○ **D.** Random access - permissions are assigned randomly

Q17 (2 marks)

A company implements a password policy requiring passwords to be changed every 90 days, with no reuse of the previous 5 passwords.
What security benefit does this provide?

- ○ **A.** It makes passwords easier to remember
- ○ **B.** It speeds up the login process
- ○ **C.** It limits the time window a compromised password remains valid and prevents cycling through the same passwords
- ○ **D.** It reduces the number of passwords users need

Q18 (2 marks)

Anti-malware software uses two main detection methods: signature-based detection and heuristic analysis.
What is the difference between these methods?

- ○ **A.** Signature detection is faster; heuristic is slower - no other difference
- ○ **B.** Signature detection compares files against known malware patterns; heuristic analysis examines behaviour to detect unknown threats
- ○ **C.** Signature detection only works on Macs; heuristic only works on Windows
- ○ **D.** They are exactly the same method with different names

Q19 (2 marks)

A data centre implements multiple physical security layers: perimeter fencing, security guards, key card access, and biometric scanners at the server room.
Why are multiple layers of physical security used?

- ○ **A.** To make the building look more impressive to visitors
- ○ **B.** Legal requirements mandate exactly four security measures

- ○ **C.** One layer is sufficient; the others are unnecessary
- ○ **D.** Defence in depth - if one security layer fails, others still protect the assets; each layer adds difficulty for attackers

Q20 (2 marks)

A company uses encryption for data 'at rest' on hard drives and data 'in transit' over networks.
Why are both types of encryption necessary?

- ○ **A.** They are the same thing - only one is needed
- ○ **B.** Data faces different threats when stored versus when transmitted; both states need protection from different attack vectors
- ○ **C.** Encryption at rest is for old data; in transit is for new data
- ○ **D.** Only transit encryption is needed; stored data is always safe

Q21 (2 marks)

An organisation uses both hardware firewalls at the network perimeter and software firewalls on individual computers.
What is the benefit of this approach?

- ○ **A.** The hardware firewall protects the network boundary while software firewalls protect individual devices, providing layered defence
- ○ **B.** It wastes resources - only one type is needed
- ○ **C.** Hardware firewalls work better on Mondays; software firewalls work better on weekends
- ○ **D.** They cancel each other out, reducing security

Q22 (2 marks)

A penetration test can be conducted as 'black box' (testers have no prior knowledge) or 'white box' (testers have full system information).
What is the advantage of black box testing?

- ○ **A.** It is always cheaper than white box testing
- ○ **B.** It tests more systems simultaneously
- ○ **C.** It simulates a real external attacker with no insider knowledge, revealing vulnerabilities visible from outside the organisation
- ○ **D.** It requires no technical skills

Q23 (2 marks)

Two-factor authentication (2FA) requires users to provide both a password and a second verification factor, such as a code sent to their phone.
Why is this more secure than passwords alone?

- ○ **A.** It is not more secure - it just takes longer to log in
- ○ **B.** Two passwords are easier to remember than one
- ○ **C.** An attacker would need to compromise both factors; knowing the password alone is not enough to gain access
- ○ **D.** Mobile phones cannot be hacked

Q24 (3 marks)

A company implements: a firewall blocking unauthorised traffic, anti-malware on all devices, encrypted data storage, user access controls, and regular penetration testing.
Analyse how these measures work together to provide comprehensive security.

- ○ **A.** They all do the same thing - only one is needed
- ○ **B.** These measures are in conflict and reduce overall security
- ○ **C.** Only penetration testing is useful; other measures have no value
- ○ **D.** Firewall blocks network threats, anti-malware catches what passes through, encryption protects stolen data, access controls limit damage from compromised accounts, and penetration testing validates all measures - each addresses different attack vectors

Q25 (3 marks)

A penetration test reveals: weak admin passwords, an unpatched server with a known vulnerability, and employees clicking links in test phishing emails. The company has limited budget for fixes.
Analyse which vulnerabilities should be prioritised and why.

- ○ **A.** Fix nothing - if systems work currently, they are secure enough
- ○ **B.** Prioritise all three: patch the server (known exploit available), enforce strong passwords (easy fix, prevents unauthorised access), and implement security training (addresses human factor) - but the unpatched server is most critical as exploits exist
- ○ **C.** Only fix password issues - the others will resolve themselves
- ○ **D.** Penetration test results should be ignored as they are not real attacks

Q26 (3 marks)

A healthcare organisation stores patient records. They implement: encryption for data at rest on servers, encryption for data in transit using HTTPS, encrypted backups stored offsite, and encrypted laptops for staff who work remotely.
Analyse how each encryption measure protects against different threats.

- ○ **A.** All four measures protect against the same threat - only one is needed
- ○ **B.** Encryption is unnecessary for healthcare data
- ○ **C.** Only laptop encryption matters; servers are always physically secure
- ○ **D.** Server encryption protects against data centre breaches; HTTPS protects against network interception; backup encryption protects against backup theft; laptop encryption protects against device theft - each addresses a specific threat scenario

Q27 (3 marks)

An organisation suffers a data breach. Investigation reveals an employee account was compromised through phishing. The attacker accessed HR records, financial systems, and customer databases - all using this single account.
Analyse what access control failures contributed to the breach scope.

- ○ **A.** Access controls worked correctly - the breach was unavoidable
- ○ **B.** The account had excessive privileges violating least privilege; no separation of duties existed between HR, finance, and customer data; access reviews likely failed to identify unnecessary permissions
- ○ **C.** More passwords would have prevented the breach
- ○ **D.** Physical security failures caused this breach

Q28 (3 marks)

Two companies protect their servers differently. Company A uses only digital security (firewalls, encryption, access controls). Company B adds physical security (locked server room, CCTV, security guards).
Analyse why Company B's approach provides better protection.

- ○ **A.** Both approaches are equivalent - physical security is unnecessary in modern computing
- ○ **B.** Physical security is more important than digital security
- ○ **C.** Company A's approach is better because digital security is sufficient
- ○ **D.** Physical security prevents direct hardware access, protects against insider theft, deters social engineering for physical access, and ensures digital controls cannot be bypassed by physical tampering - digital-only security leaves physical vulnerabilities

Q29 (3 marks)

Despite having updated anti-malware software, a company's network becomes infected with ransomware. The malware was a new variant that had never been seen before.
Analyse why anti-malware failed and what additional measures could help.

- ○ **A.** Anti-malware software is useless and should be removed
- ○ **B.** The anti-malware was definitely out of date despite claims otherwise
- ○ **C.** Nothing could have prevented this attack
- ○ **D.** The zero-day malware had no existing signature; additional measures like behaviour-based detection, network segmentation to limit spread, regular backups for recovery, and user training to prevent initial infection would provide defence in depth

Q30 (3 marks)

A school implements comprehensive cyber security: network firewall, anti-malware on all devices, complex password policy, access controls

by role, encrypted Wi-Fi, CCTV and locked IT areas, plus regular staff training. Yet students still attempt to bypass restrictions.
Analyse why technical and physical measures alone cannot guarantee security.

- ○ **A.** The school should remove all security measures since they do not work
- ○ **B.** Students are too clever to secure against
- ○ **C.** Only punishment, not security measures, can prevent breaches
- ○ **D.** Security requires ongoing effort: determined users find workarounds, new vulnerabilities emerge constantly, human behaviour is unpredictable, and security measures must evolve - the human factor means perfect security is impossible

Total: **55** marks • Time: **30 min**

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